# Scalable Semantic Search Using Vector Embeddings

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## Abstract

Modern information retrieval systems increasingly rely on dense semantic representations rather than traditional keyword matching. We investigate the efficiency and recall tradeoffs in high-dimensional vector spaces using approximate nearest neighbor algorithms. Experimental evaluations across ten million documents demonstrate a 4x throughput improvement while preserving over 96% top-k recall accuracy. Furthermore, we analyze index compression techniques that reduce memory footprints by 60%.

**Keywords: Information Retrieval, Vector Search, Machine Learning, Deep Learning, Database Systems**

## 1. Introduction

Traditional lexical search methods such as BM25 remain pervasive due to their inverted index efficiency. However, vocabulary mismatch and synonymy severely constrain retrieval performance in specialized academic literature. Dense retrieval using learned embeddings bridges this semantic gap by projecting queries and documents into a shared metric space.

## 2. Vector Compression & Indexing

We analyze product quantization (PQ) alongside hierarchical navigable small world (HNSW) graphs. By quantizing sub-vectors into codebooks, we retain high recall while dramatically diminishing memory bandwidth pressure on modern multi-core servers.

## 3. Empirical Findings

Throughput measurements on standardized multi-million passage corpora demonstrate query latencies under 12 milliseconds at 99th percentile, facilitating real-time interactive search across expansive research libraries.

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